

CLASS :
EXAM NO :

TIME : 3 Hrs
MARKS : 80

Class 11 Biology

Unit: 4

Difficulty: Hard

Total Marks: 100

Duration: 3 hours

SECTION A: MCQs (1 mark each) (14 × 1 = 14)

- 1) Which of the following characteristics distinguishes ctenophores from cnidarians?
 - a) Radial symmetry
 - b) Diploblastic organization
 - c) Presence of comb plates
 - d) Intracellular digestion
- 2) In ctenophores, the comb plates are primarily used for:
 - a) Feeding
 - b) Digestion
 - c) Locomotion
 - d) Excretion
- 3) Assertion: Ctenophores are exclusively marine organisms.
Reason: Ctenophores cannot regulate their osmotic balance in freshwater environments.
 - a) Both assertion and reason are true, and the reason is the correct explanation
 - b) Both assertion and reason are true, but the reason is not the correct explanation
 - c) Assertion is true but reason is false
 - d) Both assertion and reason are false
- 4) Which of the following features is NOT characteristic of phylum Ctenophora?
 - a) Tissue level of organization
 - b) Bioluminescence
 - c) Hermaphroditism
 - d) Exoskeleton
- 5) Based on the NCERT text, which of these statements about ctenophores is incorrect?
 - a) They possess eight rows of ciliated comb plates
 - b) They exhibit both extracellular and intracellular digestion
 - c) They have separate sexes

- d) They are commonly known as sea walnuts
- 6) Ctenophores are classified under which level of organization?
- a) Organ-system level
 - b) Tissue level
 - c) Cellular level
 - d) Colonial level
- 7) A student observed a marine organism with eight rows of ciliated plates that help in movement. The organism is most likely:
- a) A sea anemone
 - b) A comb jelly
 - c) A sponge
 - d) A flatworm
- 8) Which of the following is a unique feature of ctenophores that helps in their identification?
- a) Presence of nematocysts
 - b) Bioluminescence
 - c) Bilateral symmetry
 - d) Presence of tentacles
- 9) Case study: A marine biologist collected transparent, gelatinous organisms from coastal waters. These organisms showed bioluminescence and had eight rows of cilia. They were most likely:
- a) Jellyfish from phylum Cnidaria
 - b) Comb jellies from phylum Ctenophora
 - c) Marine flatworms from phylum Platyhelminthes
 - d) Marine annelids from phylum Annelida
- 10) Assertion: Ctenophores exhibit both intracellular and extracellular digestion.
Reason: They have a gastrovascular cavity where food is partially digested before being taken up by cells.
- a) Both assertion and reason are true, and the reason is the correct explanation
 - b) Both assertion and reason are true, but the reason is not the correct explanation
 - c) Assertion is true but reason is false
 - d) Both assertion and reason are false
- 11) Which feature distinguishes ctenophores from poriferans?
- a) Multicellularity
 - b) Presence of specialized cells
 - c) Tissue level of organization
 - d) Marine habitat
- 12) In the evolutionary sequence of animals, ctenophores represent advancement over cnidarians because:
- a) They have a complete digestive system
 - b) They have a more complex nervous system
 - c) They have both intracellular and extracellular digestion
 - d) They have specialized locomotory structures
- 13) Which of the following is true about reproduction in ctenophores?

- a) They reproduce only asexually
- b) They have separate sexes
- c) They are hermaphroditic
- d) They reproduce through binary fission

14) A student studying marine organisms collected specimens with the following characteristics: radially symmetrical, diploblastic, tissue-level organization, and eight rows of ciliated comb plates. Based on these observations, the organisms belong to:

- a) Phylum Porifera
- b) Phylum Cnidaria
- c) Phylum Ctenophora
- d) Phylum Platyhelminthes

SECTION B: Very Short Answer (2 marks each) (10 × 2 = 20)

- 15) Explain how the floral diagram of a zygomorphic flower would differ from an actinomorphic flower, with reference to the arrangement of floral parts.
- 16) Compare the structure of vascular bundles in monocot and dicot stems, highlighting key anatomical differences that allow for their identification.
- 17) Analyze how the arrangement of tissues in a dorsiventral leaf contributes to its photosynthetic efficiency compared to an isobilateral leaf.
- 18) Differentiate between the structure and function of bulliform cells in monocot leaves and explain their adaptive significance during water stress.
- 19) Describe how the anatomical features of endodermis in roots contribute to selective water and mineral absorption.
- 20) Explain how the internal structure of a dicot stem facilitates secondary growth, while a monocot stem generally lacks this capability.
- 21) Compare the position and arrangement of sclerenchyma in monocot and dicot stems, and relate this to their respective mechanical support requirements.
- 22) How does the structure of guard cells in grasses differ from those in dicots, and what functional advantage might this difference provide?
- 23) Explain the anatomical basis for the difference in venation patterns between monocot and dicot leaves, with reference to the arrangement of vascular bundles.
- 24) Analyze how the position of stomata differs between the adaxial and abaxial surfaces of a dorsiventral leaf, and explain the ecological significance of this arrangement.

SECTION C: Short Answer Type I (3 marks each) (7 × 3 = 21)

- 25) Explain how the internal structure of a dicot stem is adapted for efficient transport of water and minerals. How does this differ from the arrangement in monocot stems?
- 26) Analyze the functional significance of the bulliform cells in grass leaves. How do these specialized cells help the plant adapt to different environmental conditions?
- 27) Compare and contrast the anatomy of dicot and monocot roots, focusing on the arrangement of vascular tissues. How do these differences reflect their evolutionary adaptations?

- 28) Examine how the internal structure of dorsiventral leaves facilitates efficient photosynthesis. What specific anatomical features contribute to this efficiency?
- 29) Describe how the respiratory system of frogs is adapted for both aquatic and terrestrial environments. What structural features enable this dual mode of respiration?
- 30) Analyze the significance of sexual dimorphism in frogs. How do these morphological differences relate to their reproductive behavior?
- 31) Evaluate the ecological importance of frogs in maintaining ecosystem balance. How does their anatomy and physiology contribute to their role in food chains?

SECTION D: Case-based/Competency (4 marks each) ($2 \times 4 = 8$)

- 32) 1 Based on the characteristics described, identify the phylum to which this organism belongs. (1 mark)
- 33) 2 How does the locomotion mechanism in this organism differ from that found in cnidarians? (1 mark)

SECTION E: Long Answer (5 marks each) ($3 \times 5 = 15$)

Long Answer Questions (5 marks each) with Internal Choice

- 1) Describe the phylum Ctenophora with reference to its distinctive characteristics, locomotion mechanism, and bioluminescence. Compare and contrast ctenophores with cnidarians in terms of their body organization and digestive processes.

OR

Explain the structure and function of the tissue systems in flowering plants. How do the epidermal, ground, and vascular tissue systems contribute to the overall functioning of the plant body? Discuss the variations in these tissue systems between monocots and dicots.

- 2) Explain the structure and functioning of the respiratory system in frogs, describing how they are adapted for both aquatic and terrestrial respiration. Discuss the significance of cutaneous respiration in amphibians with reference to their evolutionary adaptations.

OR

Describe the anatomy of a dorsiventral (dicotyledonous) leaf with labeled diagrams. Explain how the internal structure of the leaf is adapted for its primary function of photosynthesis. How does this structure differ from that of a monocotyledonous leaf?

- 3) Discuss the morphology and anatomy of the flower as a reproductive unit in angiosperms. Explain the significance of different floral parts in sexual reproduction and how these are represented in floral diagrams and floral formulae. Illustrate your answer with suitable examples.

OR

Explain the excretory and reproductive systems of frogs. How are these systems structurally and functionally interconnected in male frogs but separate in females? Discuss the evolutionary significance of this arrangement in amphibians.

QUESTION PAPER GENERATION SUMMARY

Class: 11

Subject: Biology

Unit: 4

Difficulty: Hard

Total Marks: 100

Duration: 3 hours

Section Details:

Section A (MCQs): 14 questions | 1 mark each | Total: 14 marks

Chapters: 4, 5 | Context: 27643 chars

Section B (Very Short Answer): 10 questions | 2 mark each | Total: 20 marks

Chapters: 5, 6 | Context: 26654 chars

Section C (Short Answer Type I): 7 questions | 3 mark each | Total: 21 marks

Chapters: 6, 7 | Context: 27717 chars

Section D (Case-based/Competency): 2 questions | 4 mark each | Total: 8 marks

Chapters: 4, 5, 6, 7 | Context: 55361 chars

Section E (Long Answer): 3 questions | 5 mark each | Total: 15 marks

Chapters: 4, 5, 6, 7 | Context: 55361 chars